

Changyu (Thomason) Zhao

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EDUCATION

Stanford University

M.S. in Computer Science (Computer and Network Security)

GPA: **4.0/4.0**

University of Wisconsin-Madison

B.S. in Computer Science (Honors) and Mathematics

GPA: **3.95/4.0**, David Dewitt Undergraduate Scholarship, Dean's List (all semesters)

Sep. 2025 – Jun. 2027 (expected)

Stanford, California

Sep. 2021 – May 2025

Madison, Wisconsin

EXPERIENCES

UW-Madison Cybersecurity Operation Center

Cybersecurity Student Intern Analyst

Feb. 2023 – May 2025

Madison, Wisconsin

- Investigated **1000+** security incidents (malware, phishing, intrusions) using **Elastic SIEM**. Developed **threat hunting queries and detection rules** that reduced MTTR by **40%** and repeat infections by **25%**.
- Conducted monthly **IAM attestations** for **500+** accounts using automated scripts, remediating 3% toxic access in 24h.

UW-Madison MadS&P Lab

Research Assistant, Advisor: Prof. Patrick McDaniel

Sep. 2024 – Aug. 2025

Madison, Wisconsin

- Conducted **comprehensive evaluation** of LibIHT against Intel Pin and DynamoRIO on real-world workloads, demonstrating **150x speedup**, **99%+ code coverage**, and **complete resistance** to anti-debugging techniques employed by malware.
- Established **reproducible benchmarking infrastructure** with 21 coreutils binaries and 10 malware specimens, automating performance metrics collection and visualization in **Python**. Paper accepted by **ACM CCS 2025 SURE** workshop.

ASU SEFCOM Lab

Research Assistant, Advisor: Prof. Yan Shoshitaishvili and Ruoyu Wang

May. 2024 – Sep. 2024

Tempe, Arizona

- Co-designed Oxidizer, the first **Rust-focused decompiler** built atop angr, implementing control-flow recovery, function prototype inference, and macro inference to address the unique challenges of analyzing memory-safe compiled binaries.
- Developed **static type recovery engine** to extract Rust-specific type metadata from compiled binaries and integrates findings into decompilation, improving accuracy of data structure and function signature reconstruction for security analysis.

Tencent Xuanwu Lab

Security Researcher Intern

Jul. 2023 – Aug. 2023

Beijing, China

- Architected **LibIHT**, a cross-platform dynamic binary tracing framework in **C/C++/Python** that leverages Intel LBR/BTS hardware features for control-flow tracing, achieving **150x speedup** over Intel Pin on large-scale program analysis.
- Designed unified **kernel driver architecture** for Linux and Windows with abstraction layers for OS and hardware, reducing code duplication by **40%** while standardizing trace session management and improving cross-platform maintainability.
- Implemented **user-space API library** and debugger plugin system for **GDB/WinDbg** integration, enabling security analysts to visualize execution traces, identify anomalous control-flow patterns, and accelerate vulnerability discovery workflows.

ASU SEFCOM Lab

Research Assistant, Advisor: Prof. Yan Shoshitaishvili

May. 2023 – Jul. 2023

Tempe, Arizona

- Developed **LibHopper**, an automated exploitation primitive discovery tool combining debugger instrumentation with **symbolic execution** to identify exploitable memory corruption opportunities in library state structures.
- Constructed **exploitation primitives dataset** spanning critical C libraries (OpenSSL, OpenSSH, libcurl, libpng), providing structured risk assessments to inform security research in automated exploit generation and vulnerability scanning tooling.

PROJECTS

Router automated vulnerability discovery | IoT, Reverse Engineering, Web Security

May 2022 – Aug. 2022

- Built an **IDA plugin** for static IoT firmware analysis, uncovering **7 unfixed CVEs** in the Tenda AC18 router.
- Performed dynamic analysis with **QEMU** rehosting and **GDB** to develop PoC exploits for identified vulnerabilities.

PRESENTATIONS & PUBLICATIONS

Yibo Liu, ..., **Changyu Zhao**, ... "Oxidizer: Toward Concise and High-fidelity Rust Decompilation". *2026 IEEE Symposium on Security and Privacy (IEEE S&P)*. May 2026.

Changyu Zhao, ... "LibIHT: A Hardware-Based Approach to Efficient and Evasion-Resilient Dynamic Binary Analysis".

Proceedings of the 1st Workshop on Software Understanding and Reverse Engineering (SURE 2025). ACM, October 2025.

Black Hat Arsenal, *LIBIHT: A Cross-Platform Library for Accessing Intel Hardware Trace Features*, Las Vegas, NV, Aug. 2024

EXTRACURRICULAR ACTIVITIES

r3kapig CTF Team

Team Member

Sep. 2021 – Present

International

- Entered **DEFCON CTF world final** from 2022 to 2025, ranked 12th, 7th, 4th, and 5th respectively.
- Competed in **120+** CTF competitions, expert in kernel and user-space exploitation and reverse engineering.

SKILLS

Languages: C/C++, Python, Rust, Go, Java, x86/x64, ARM, MIPS Assembly

Systems & Reverse Engineering: QEMU, GDB, WinDbg, IDA Pro, Ghidra

Security Analysis: Angr, AFL++, Frida, Joern, Wireshark, Metasploit, Kali